

Beer color, alkalinity and mash pH

From German brewing and more

Mash pH is the result of the balance between grist acidity and water alkalinity. The acidity of the grist is determined by the malts used and darker malts are generally more acidic than lighter colored ones. The color of the malts used in the grist also determine the beer color to a large extent. On the other hand water alkalinity, to be correct its residual alkalinity, is determined by its mineral composition. It therefore stands to reason that beer color and water composition, necessary for a proper mash pH, are related.

This article uses results from mash pH experiments

(http://braukaiser.com/documents/effect_of_water_and_grist_on_mash_pH.pdf) to shed light on the relationship between beer color, mash pH and water composition. It develops a formula that can be used to make a crude prediction of the mash pH, or the alkalinity necessary for a given mash pH, based on the color and mash thickness of the beer. This formula has been implemented in the water calculator (Kaiser_water_calculator.xls

(http://braukaiser.com/documents/Kaiser_water_calculator.xls)) to predict mash pH from beer color, mash thickness and water composition.

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Malt color, type and acidity

Brewers know that darker malts are more acidic. But what does it mean for a malt to be more acidic? They for sure don't taste sour.



Malt acidity is the malt's ability to lower mash pH and it can be measured via 2 means. One is the pH of a distilled water mash. Because of the absence of pH affecting water ions the pH of that mash is determined only by malt acidity and mash thickness. Another approach is to take a sample from such a mash and add a strong base (e.g sodium hydroxide) to it until a predetermined pH (e.g pH 5.7) is reached. This process is called titration. The amount of base added per unit of malt is a direct measure of that malt's acidity. Testing the distilled water mash pH works well for base malts. Specialty malts, however, are generally much more acidic than base malts and testing their acidity through titration works better.

In aforementioned mash pH experiments (http://braukaiser.com/documents/effect_of_water_and_grist_on_mash_pH.pdf) the following formula was developed for calculating the distilled water pH of a given grist.

$$pH_{DI\ water\ mash} = \left(\sum pH_{bi} \cdot g_{bi} + 5.7 \cdot \sum g_{sj} \right) - \frac{0.14 \cdot \sum a_{sj} \cdot g_{sj}}{R_{mash}}$$

Where:

- **pH_{DI water mash}**: the mash pH of the grist in distilled water
- **pH_{bi}**: the distilled water mash pH of the particular base malt i
- **g_{bi}**: the contribution of base malt i to the weight of the grist (between 0 and 1)
- **g_{sj}**: the contribution of the specialty malt j to the weight of the grist (between 0 and 1)

- a_{sj} : the acidity of the specialty malt j in mEq/kg
- R_{mash} : the mash thickness in l/kg

In English: the weighted average of the distilled water mash pH for all the base malts and the titration point for specialty malts (5.7) is determined. This pH is then lowered by the acidity of the specialty malts. The more acidic they are, the higher their grist percentage and the thicker the mash is the lower the mash pH of the grist will be.

Random recipe creation and pH and color calculations

Using the the measured values for the distilled water pH of selected base malts and acidity of selected specialty malts 210 recipes were simulated. The recipes were thrown together randomly with the following constraints:

- 15 recipes mixed only base malts
- 45 recipes mixed base malts with up to 15% crystal malts. 50 of these recipes used only 2-row as base malt. The other 15 used a random mix of different base malts. The percentage of crystal malts in these grists was randomly chosen between 1 and 15%. In addition to that the types and and percentages of different crystal malts were also chosen randomly
- 45 recipes mixed base malts with up to 8% crystal malts and up to 8% roasted malts. 50 of these recipes used only 2 row as base malt while the other 15 used a mix of base malts. Similar to the recipes that used only crystal malts as specialty malts these recipes here had their actual crystal and roasted malt percentages determined randomly. The same applied to the types of malts that were chosen
- 45 recipes used only roasted malts. These recipes were created similar to the crystal only recipes mentioned earlier except that they only used roasted malts as base malts.

Crystal and roasted malts were treated as distinct groups since they formed distinct clusters when their acidity was plotted over their color.

For all these random recipes the distilled water mash pH of the grist and the color of the beer was calculated. The DI water mash pH was determined for 4 different mash thicknesses 2, 3, 4, and 5 l/kg. For the color calculation it was assumed that the total grist weighed 10 pound and the cast out volume was 5 gallon.

$$Lovibond_{grist} = \frac{EBC_{grist} + 1.2}{2.65}$$

$$MCU_{beer} = \frac{Lovibond_{grist} \cdot m_{grist}}{V_{castout}}$$

$$SRM_{beer} = 1.49 \cdot MCU_{beer}^{0.686}$$

The result were 4 sets of 210 data points that represent fairly realistic combinations of beer color (SRM) and the distilled water mash pH of their grists. One set for every evaluated mash thickness (2, 3, 4 and 5 l/kg). The data for 3 l/kg is shown in Figure 1.

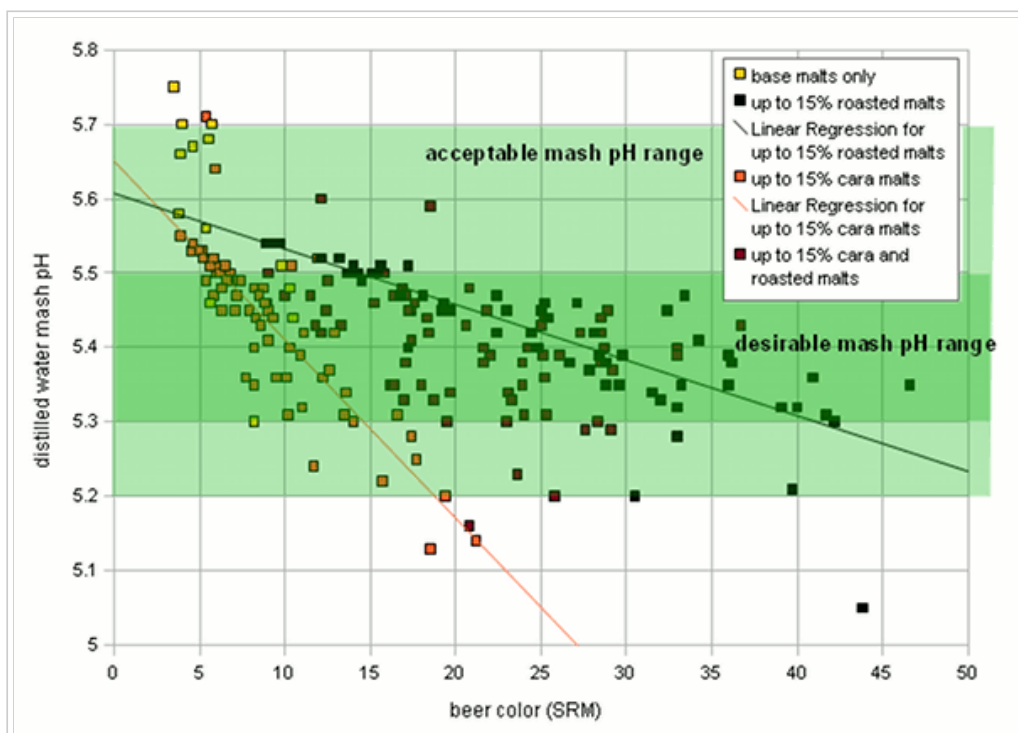


Figure 1 - distilled water mash pH over beer color for 210 randomly generated recipes

Based on this chart the following observations can be made:

- There is no simple curve that can estimate grist pH from beer color.
- a wide range of beer colors can yield an grist pH in the desired pH range and an even larger color range can yield an acceptable mash pH (Note that this does not yet take residual alkalinity of the water into account)
- "Crystal only" and "Roasted only" recipes form clusters which can be approximated reasonably well with linear functions.
- The more roasted malt that is used to achieve the desired beer color the less pH drop can be expected per unit of color. This stems from the fact that roasted malts have less acidity per unit of color compared to crystal malts

Figure 2 shows how the SRM to pH relationship changes for different water to grist ratios: the grouping remains the same, only the parameter for the linear functions change.

With these observations the idea was born to estimate the grist pH from color, mash thickness and percentage of roasted malts in the specialty malt portion of the grist. I.e. if there are no specialty malts the linear function for "cara only" is used and if all specialty malts are roasted malts the linear function of "roasted only" is used. If roasted and crystal malts are part of the grist a function that lies between the two is used.

Using the standard notation of a linear function the pH estimation from beer color and mash thickness can be written as

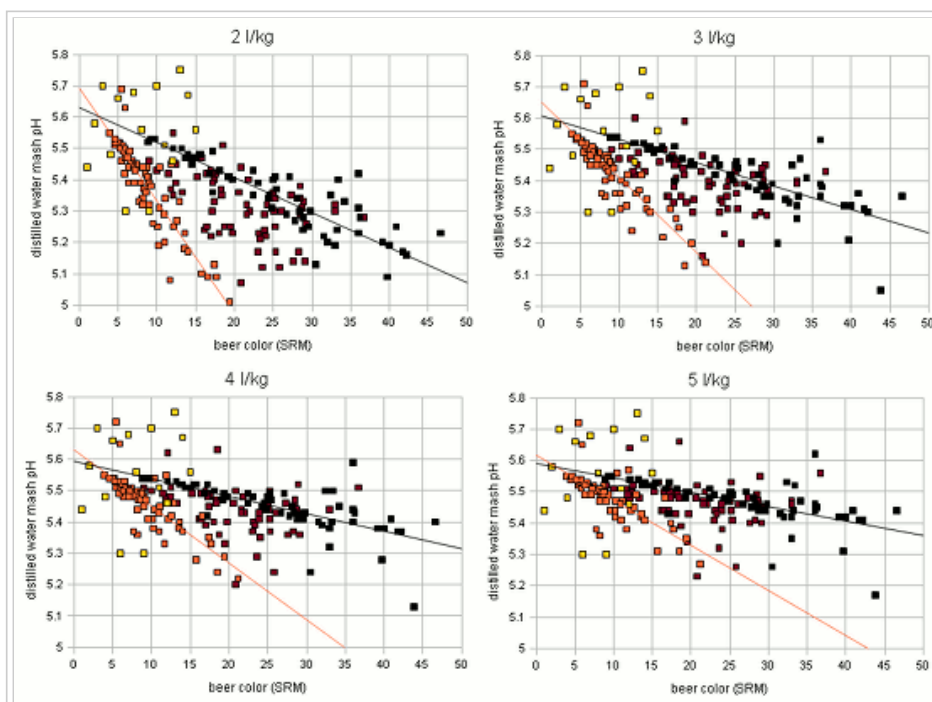


Figure 2 - Grist pH over SRM simulation of 210 different Random recipe for 4 different mash thicknesses

$$pH_{grist}(SRM_{beer}, R_{mash}, p_{roasted}) = m(R_{mash}, p_{roasted}) \cdot SRM_{beer} + b(R_{mash}, p_{roasted})$$

Where the new variable are:

- **m(R_{mash},P_{roasted})**: slope of the function which itself is a function of mash thickness and roasted malt percentage
- **b(R_{mash},P_{roasted})**: y-intercept of the function which itself is a function of mash thickness and roasted malt percentage
- **P_{roasted}**: the percentage of roasted malts in the specialty malt portion of the grist (0 - 100%)

As mentioned earlier the slope and y-intercept of the linear function are determined by simple interpolation between the slope and y-intercept for "cara only" and "roasted only" recipes:

$$m(R_{mash}, p_{roasted}) = \left(1 - \frac{p_{roasted}}{100}\right) \cdot m_{cara}(R_{mash}) + \frac{p_{roasted}}{100} \cdot m_{roasted}(R_{mash})$$

$$b(R_{mash}, p_{roasted}) = \left(1 - \frac{p_{roasted}}{100}\right) \cdot b_{cara}(R_{mash}) + \frac{p_{roasted}}{100} \cdot b_{roasted}(R_{mash})$$

Now for some more regression analysis. If the slopes and y-intercepts for "cara only" and "roasted only" recipes are plotted for the 4 different mash thicknesses (Figure 3) it is apparent that they can be fit well using a logarithmic regression.

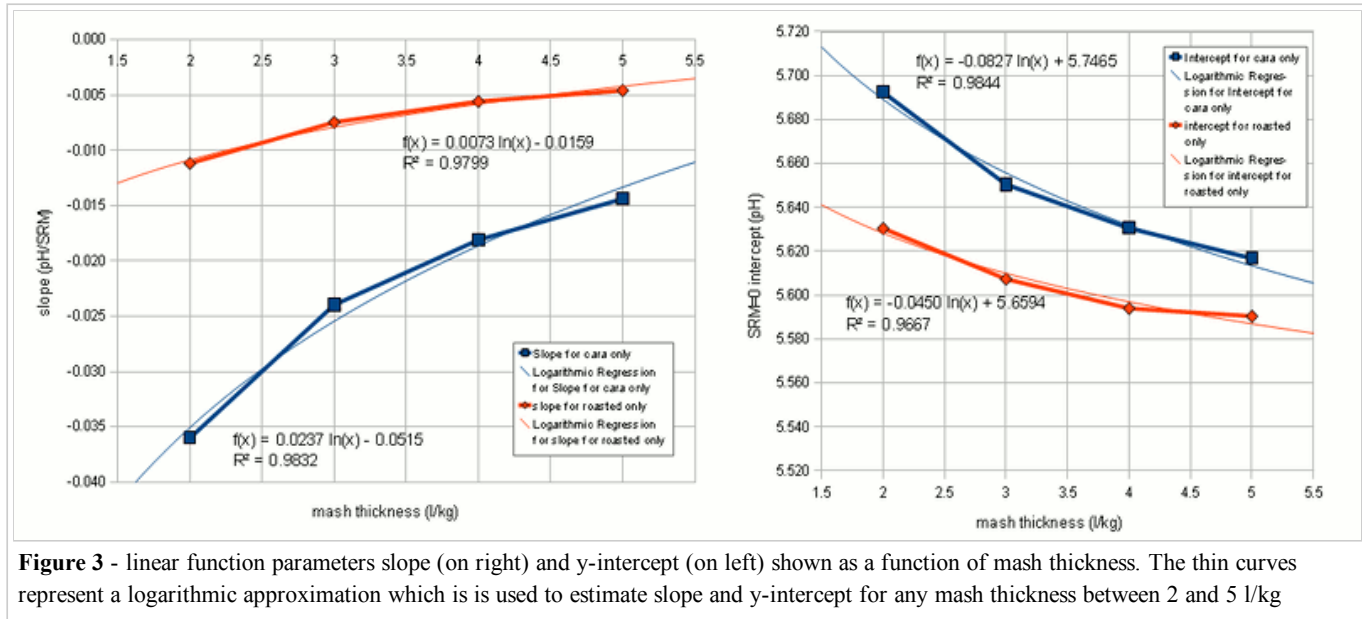


Figure 3 - linear function parameters slope (on right) and y-intercept (on left) shown as a function of mash thickness. The thin curves represent a logarithmic approximation which is used to estimate slope and y-intercept for any mash thickness between 2 and 5 l/kg

The parameters, which were found, led to the following formulas for the "cara only" and "roasted only" SRM to grist pH formula parameters slope (m) and y-intercept (b):

$$m_{cara}(R_{mash}) = 0.024 \cdot \ln(R_{mash}) - 0.052$$

$$b_{cara}(R_{mash}) = -0.083 \cdot \ln(R_{mash}) + 5.75$$

$$m_{roasted}(R_{mash}) = 0.0073 \cdot \ln(R_{mash}) - 0.016$$

$$b_{roasted}(R_{mash}) = -0.045 \cdot \ln(R_{mash}) + 5.66$$

considering residual alkalinity

Now that a method of calculating the acidity of the grist has been found, the residual alkalinity of the water (RA) needs to be considered to estimate the mash pH that a particular grist will give with the available brewing water. Mash pH experiments showed that the pH shift caused by the water's residual alkalinity depends on the residual alkalinity as well as the mash thickness.

$$\Delta pH = s_{pH} \cdot RA$$

$$s_{pH} = 0.013 \cdot R_{mash} + 0.013$$

Where:

- **RA**: residual alkalinity in mEq/l
- **s_{pH}**: pH change per 1 mEq/l residual alkalinity change

This pH shift (Delta pH) is added to the distilled water pH of the grist to predict the actual mash pH

simple Guidelines for beer color and suitable brewing waters



Using the aforementioned formulas simple guidelines for suitable brewing waters can be derived (Tables 1 and 2). These tables assume that a pH between 5.3 and 5.5 is targeted and show the residual alkalinity range for suitable brewing water based on the color of the beer and the roasted malt percentage of the specialty malt portion of the grist.

It is apparent that for any given color there is a wide range of residual alkalinities that are likely to yield a mash pH in in preferred range of 5.3 - 5.5 which also explains why most brewers do just fine with their average water when brewing moderately colored beers.

suggested residual alkalinity ranges in ppm as CaCO ₃ for brewing water					
beer color SRM	roasted malt portion of specialty malts				
	0 %	25 %	50 %	75 %	100 %
2	-300 ... -110	-300 ... -110	-290 ... -100	-290 ... -100	-290 ... -100
4	-250 ... -60	-260 ... -60	-260 ... -70	-270 ... -80	-270 ... -80
6	-200 ... -10	-220 ... -20	-230 ... -40	-240 ... -50	-260 ... -70
8	-150 ... 40	-180 ... 20	-200 ... -10	-220 ... -30	-240 ... -50
10	-100 ... 90	-130 ... 60	-170 ... 30	-200 ... 0	-230 ... -30
12	-50 ... 140	-90 ... 100	-130 ... 60	-170 ... 20	-210 ... -20
14	0 ... 190	-50 ... 140	-100 ... 90	-150 ... 40	-200 ... 0
16	40 ... 240	-10 ... 180	-70 ... 120	-120 ... 70	-180 ... 10
18	90 ... 290	30 ... 220	-40 ... 160	-100 ... 90	-170 ... 30
20	140 ... 340	70 ... 260	0 ... 190	-80 ... 120	-150 ... 40
22	190 ... 380	110 ... 300	30 ... 220	-50 ... 140	-130 ... 60
24	240 ... 430	150 ... 340	60 ... 250	-30 ... 160	-120 ... 70
26	290 ... 480	190 ... 380	90 ... 290	-10 ... 190	-100 ... 90
28	340 ... 530	230 ... 430	130 ... 320	20 ... 210	-90 ... 100
30	390 ... 580	270 ... 470	160 ... 350	40 ... 230	-70 ... 120
32	440 ... 630	310 ... 510	190 ... 380	70 ... 260	-60 ... 130
34	490 ... 680	360 ... 550	220 ... 410	90 ... 280	-40 ... 150
36	540 ... 730	400 ... 590	250 ... 450	110 ... 310	-30 ... 160
38	590 ... 780	440 ... 630	290 ... 480	140 ... 330	-10 ... 180
40	640 ... 830	480 ... 670	320 ... 510	160 ... 350	0 ... 200

mash thickness 3 l/kg
 1.4 qt/lb

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Table 1 - Suggested brewing water residual alkalinity ranges in ppm as CaCO₃ for 3 l/kg (1.4 qt/lb) mashes. This data was calculated using the formulas shown in this article with the assumption of a low end mash pH of 5.3 and a high end mash pH of 5.5

suggested residual alkalinity ranges in ppm as CaCO_3 for brewing water

beer color SRM	roasted malt portion of specialty malts				
	0 %	25 %	50 %	75 %	100 %
2	-190 ... -60	-190 ... -60	-190 ... -60	-190 ... -60	-180 ... -60
4	-170 ... -50	-170 ... -50	-180 ... -50	-180 ... -50	-180 ... -50
6	-160 ... -30	-160 ... -30	-160 ... -40	-170 ... -40	-170 ... -40
8	-140 ... -10	-150 ... -20	-150 ... -30	-160 ... -30	-170 ... -40
10	-120 ... 10	-130 ... 0	-140 ... -10	-150 ... -20	-160 ... -30
12	-100 ... 20	-120 ... 10	-130 ... 0	-140 ... -20	-160 ... -30
14	-90 ... 40	-100 ... 20	-120 ... 10	-140 ... -10	-150 ... -20
16	-70 ... 60	-90 ... 40	-110 ... 20	-130 ... 0	-150 ... -20
18	-50 ... 70	-80 ... 50	-100 ... 30	-120 ... 10	-140 ... -10
20	-40 ... 90	-60 ... 70	-90 ... 40	-110 ... 20	-130 ... -10
22	-20 ... 110	-50 ... 80	-70 ... 50	-100 ... 30	-130 ... 0
24	0 ... 130	-30 ... 100	-60 ... 70	-90 ... 30	-120 ... 0
26	20 ... 140	-20 ... 110	-50 ... 80	-90 ... 40	-120 ... 10
28	30 ... 160	0 ... 120	-40 ... 90	-80 ... 50	-110 ... 20
30	50 ... 180	10 ... 140	-30 ... 100	-70 ... 60	-110 ... 20
32	70 ... 190	20 ... 150	-20 ... 110	-60 ... 70	-100 ... 30
34	80 ... 210	40 ... 170	-10 ... 120	-50 ... 80	-100 ... 30
36	100 ... 230	50 ... 180	0 ... 130	-40 ... 80	-90 ... 40
38	120 ... 250	70 ... 200	20 ... 140	-30 ... 90	-90 ... 40
40	140 ... 260	80 ... 210	30 ... 160	-30 ... 100	-80 ... 50

mash thickness

5 l/kg

2.4 qt/lb

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Table 2 - Suggested brewing water residual alkalinity ranges in ppm as CaCO_3 for 5 l/kg (2.4 qt/lb) mashes. This data was calculated using the formulas shown in this article with the assumption of a low end mash pH of 5.3 and a high end mash pH of 5.5

final remarks

It should be noted that the approach outlined here only provides for a crude estimation of the mash pH and that there are cases where this prediction will not be correct. In particular when the distilled water mash pH of the base malts differ significantly from the pH values used in the simulation. Another limitation is the range of mash thicknesses. But it is assumed that the range of 2-5 l/kg should cover most practical mashes.

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